

GPTO paper program figures

Python reproduction of Smith and Norato (2020), Figures 4-18

Red explicit bars use α^2 opacity, matching plot_design.m.

MBB uses penalized density; L-bracket and 3D use unpenalized combined density.

The 3D surface follows Cell Data to Point Data and a $\rho = 0.5$ threshold.

MBB and L-bracket histories are fresh Python runs; exact state provenance is noted per page.

Source: <https://doi.org/10.1007/s00158-020-02552-0>

MATLAB commit: ead7250e007d4c185de59f8eee6b88a35be39550

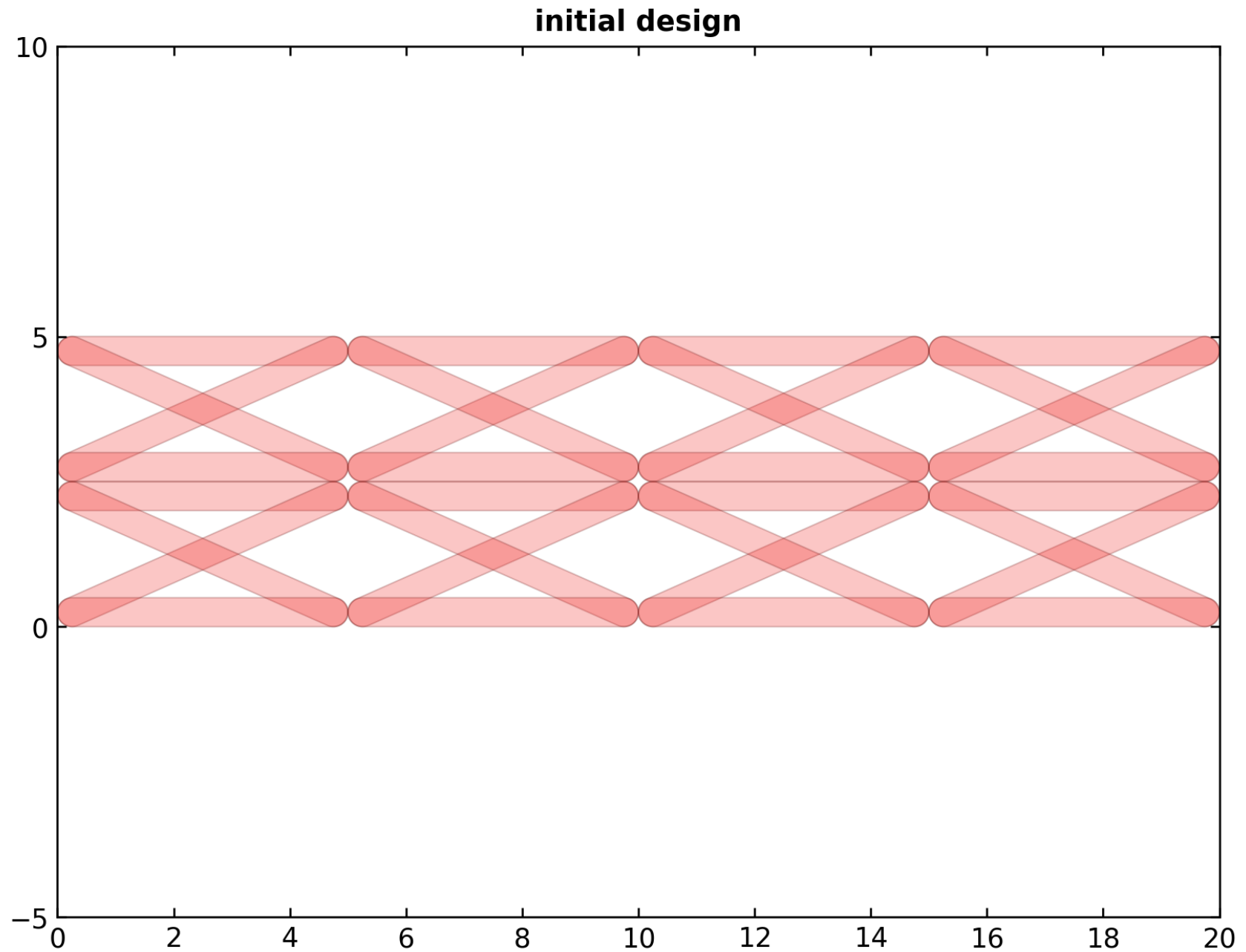
PyGPTO commit: 8157ca2f5b82feb4a2032fd4501c9f49a58b7e8c

Figure 4. MBB beam problem



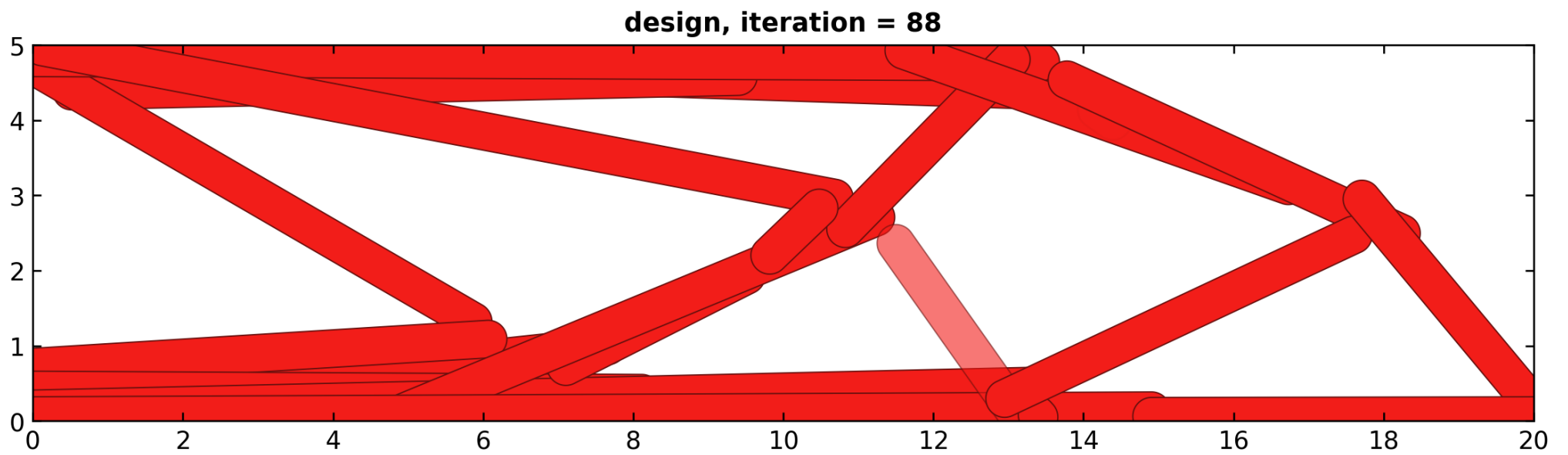
Schematic, not a numerical program output; recreated from the paper specification.

Figure 5. Initial design for MBB beam



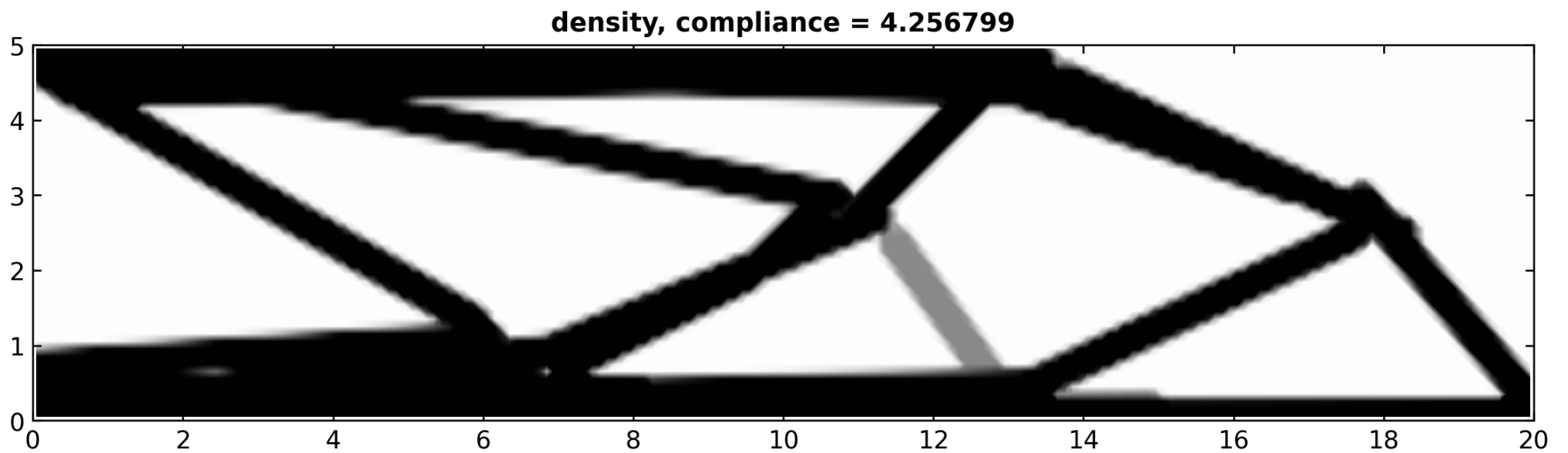
Paper viewport and code-level α^2 opacity are reproduced.

Figure 6. Optimal design for MBB beam



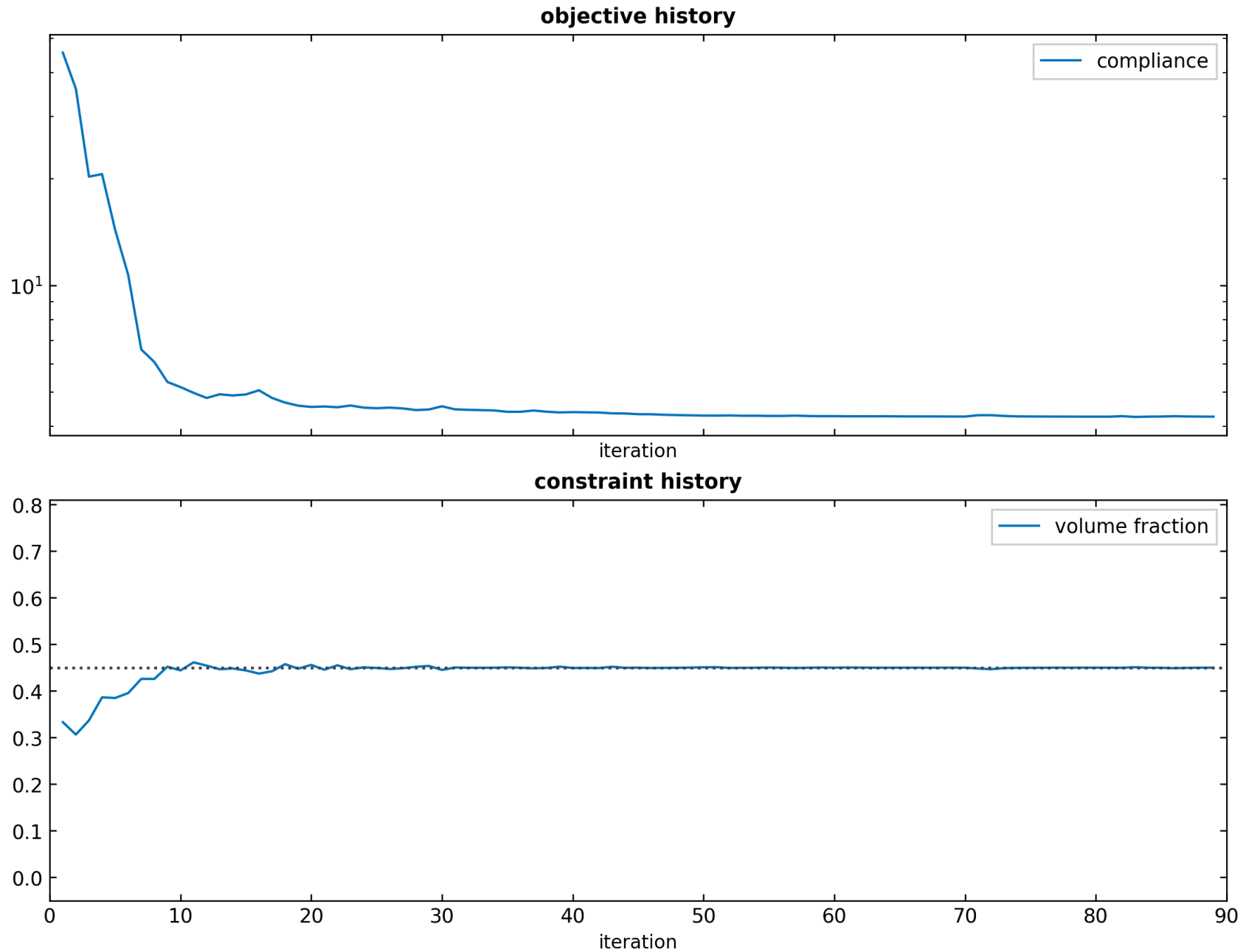
Fresh Python paper-mesh run, 88 iterations; geometry is not the unavailable exact paper checkpoint.

Figure 7. Combined density of optimal design for MBB beam



Uses penalized/effective density exactly as the released structured-mesh MATLAB plotting branch; computed compliance is 1.33% above the paper value.

Figure 8. Optimization history for MBB beam



Fresh Python run with the paper mesh and 88 optimizer iterations; semilog compliance and the 0.45 volume limit mirror plot_history.m.

Figure 9. 2D L-bracket problem

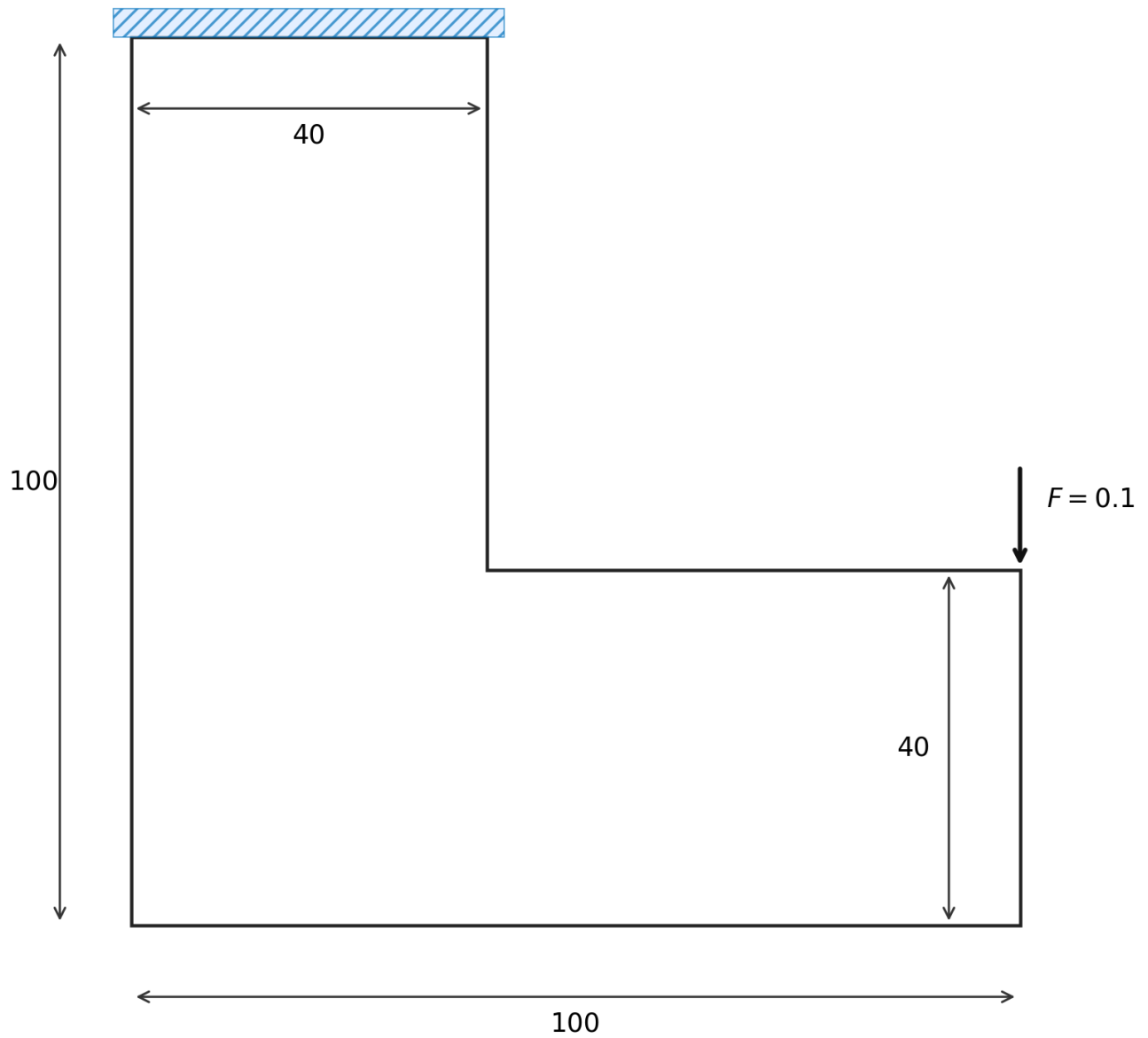
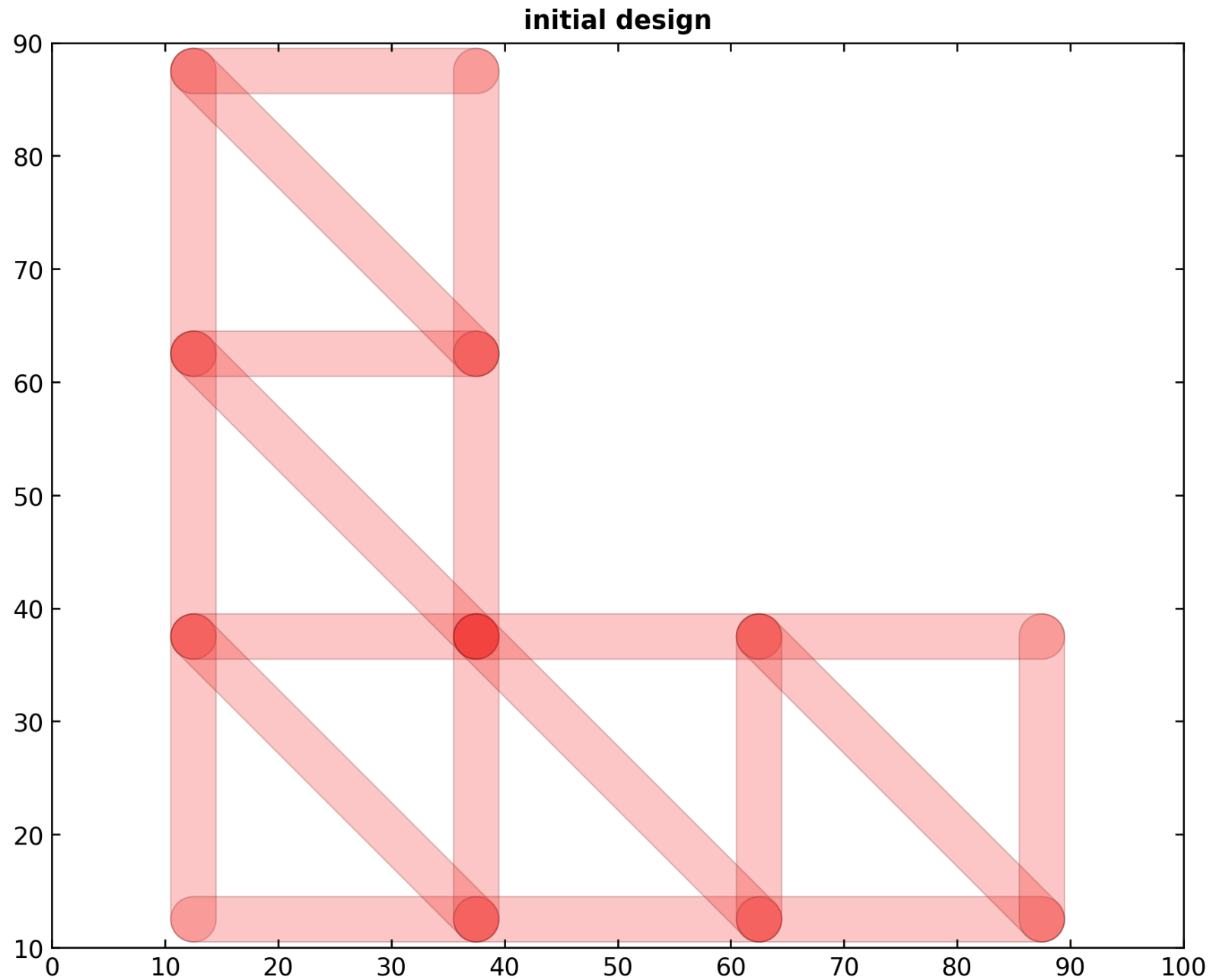
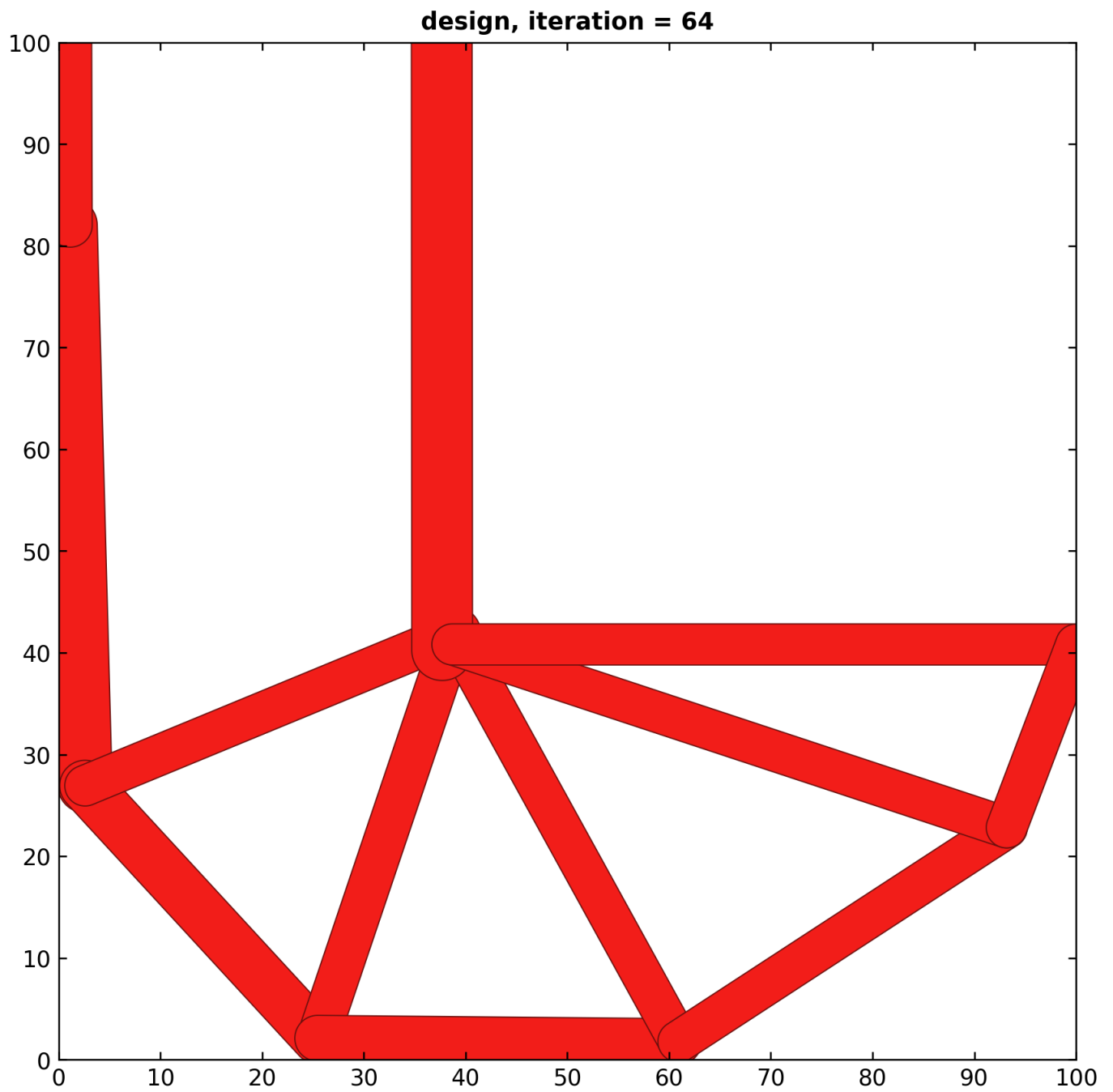


Figure 10. Initial design for L-bracket



Paper viewport and code-level α^2 opacity are reproduced.

Figure 11. Optimal design for L-bracket



Released 64-iteration final geometry; the recomputed compliance matches the paper within $7.6\text{e-}8$ relative error.

Figure 12. Combined density of optimal design for L-bracket

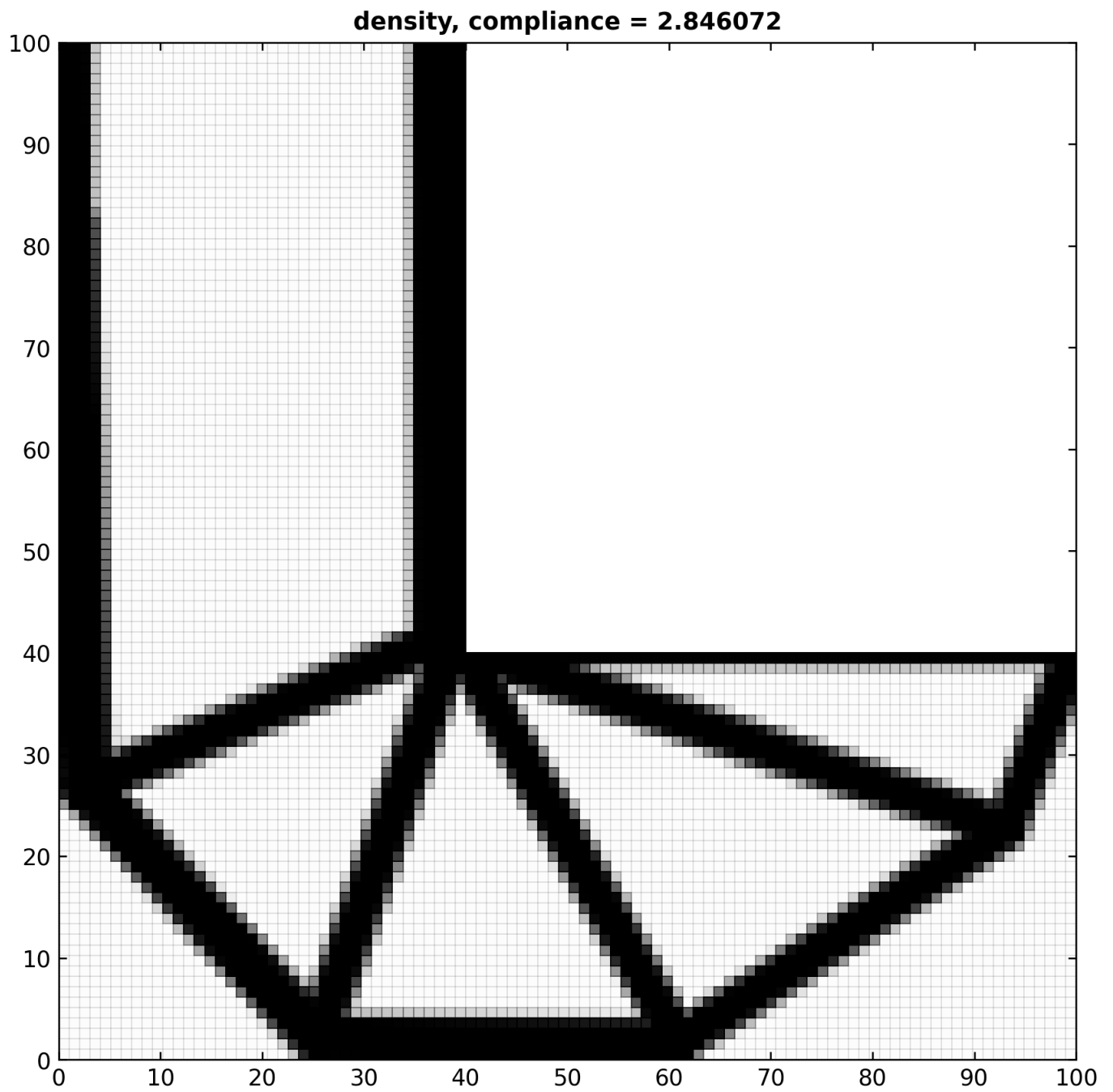
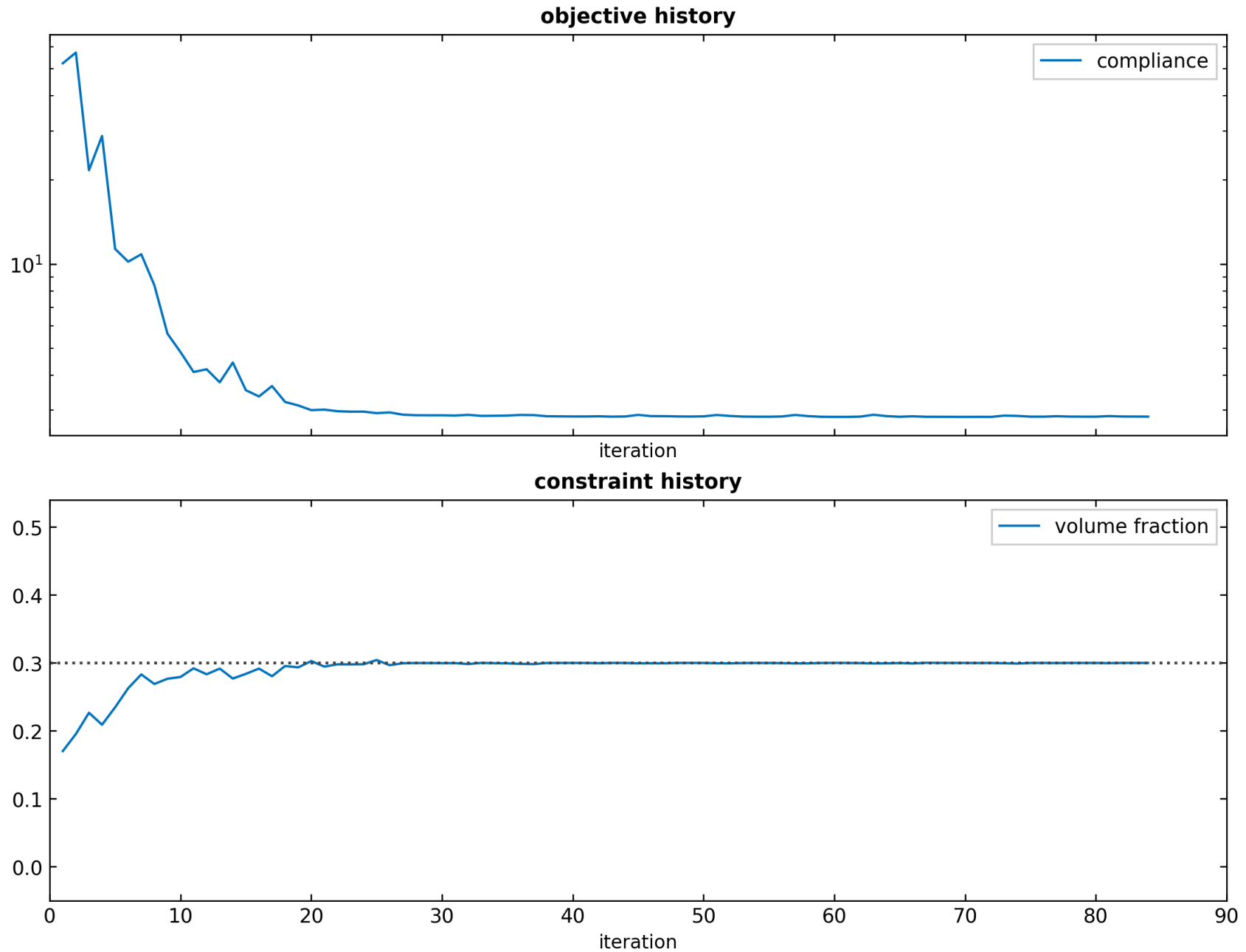
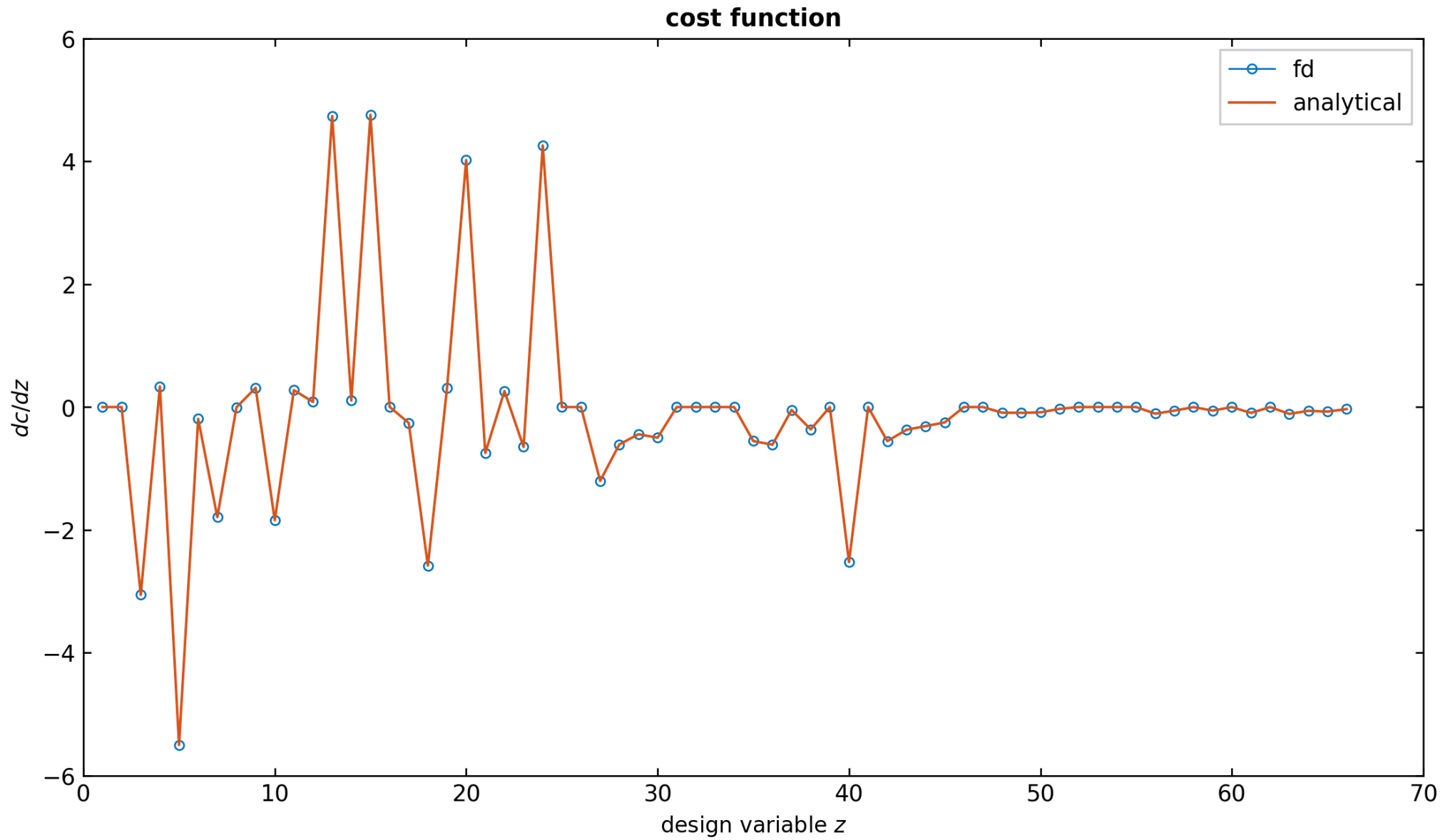


Figure 13. Optimization history for L-bracket



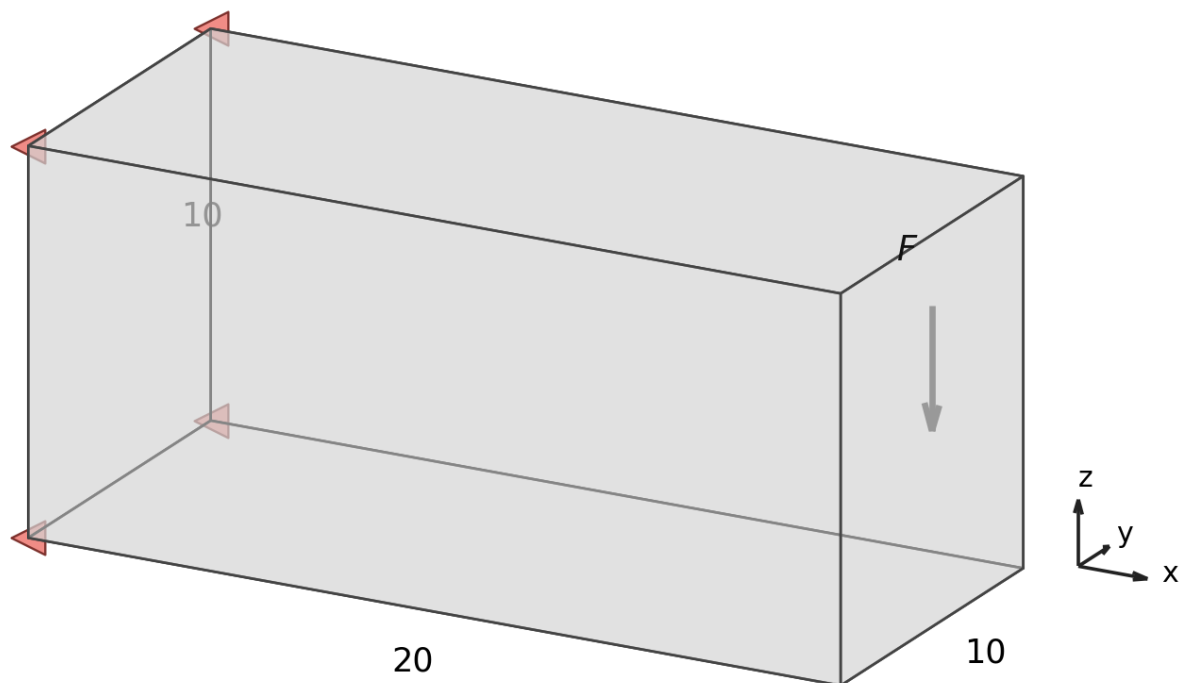
Fresh Python paper-mesh history (83 optimizer iterations); the exact 64-step MATLAB history was not published in either repository.

Figure 14. Finite-difference check of L-bracket compliance sensitivities



All 66 scaled variables use a one-sided forward difference with $h=1e-6$; the maximum signed discrepancy reproduces the paper's -0.0037 and -0.0013 values.

Figure 15. 3D cantilever problem



The paper caption calls this an initial design, but the body text and image show the load/support schematic; the true initial geometry is Fig. 16.

Figure 16. Initial design for 3D cantilever beam

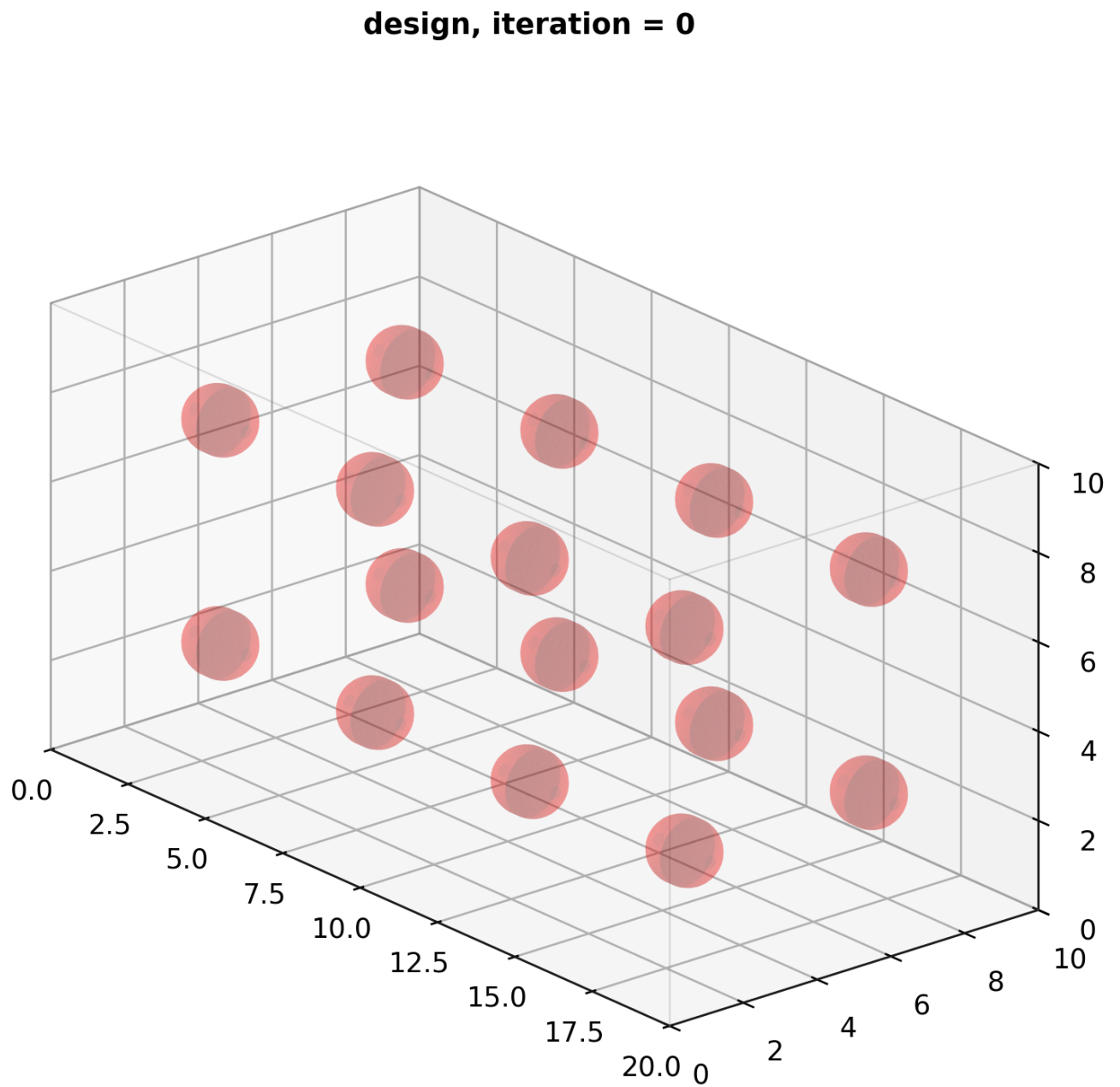


Figure 17. Optimal design for 3D cantilever beam

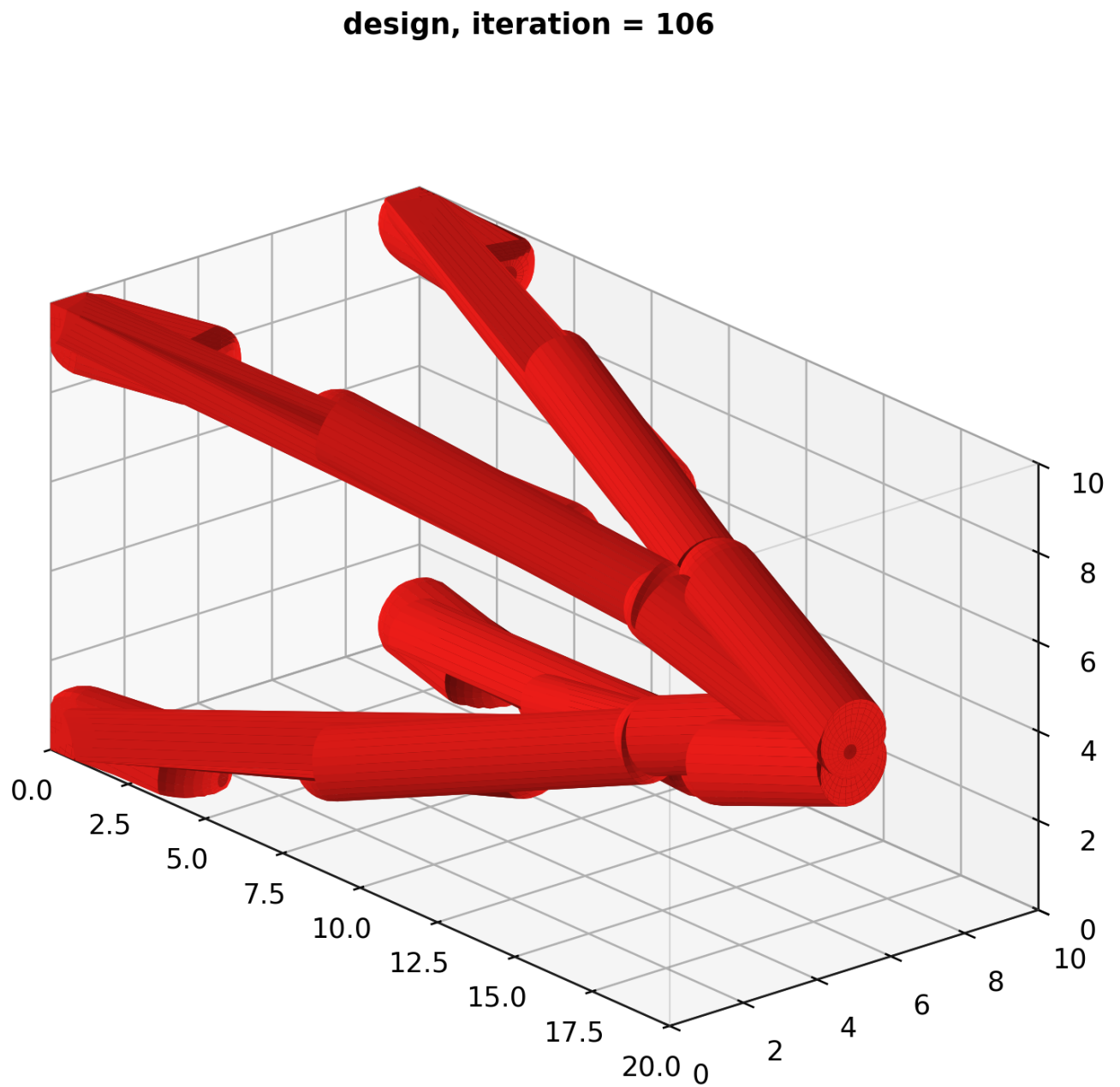
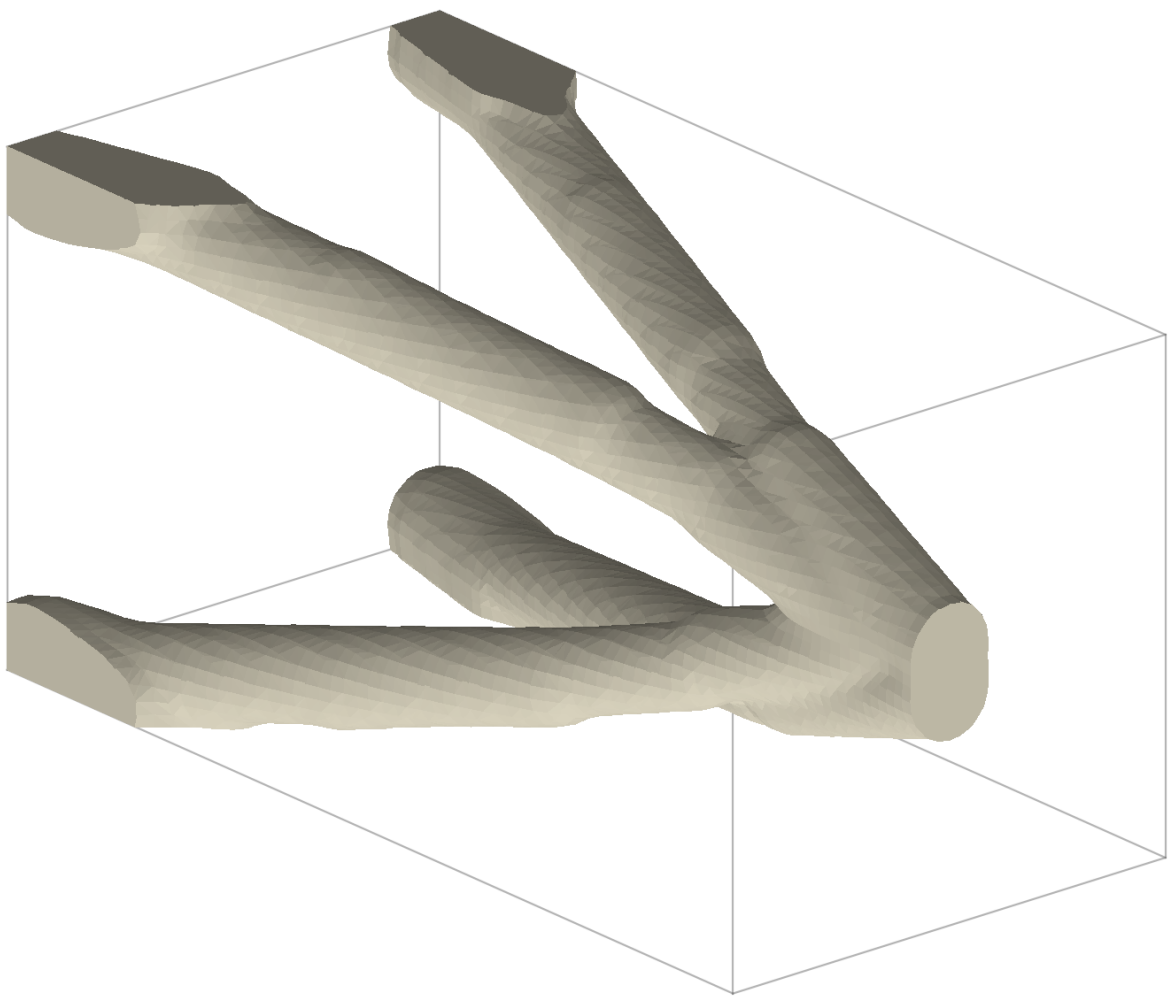


Figure 18. Combined density isosurface for optimal 3D cantilever beam



Unpenalized combined density, Cell Data to Point Data averaging, and $\rho=0.5$ are reproduced; the unpublished ParaView camera and lighting are visually approximated.